1 · Source

knowledge_base/ · 7 docs, 4.2k words

topology/ip_device_map.md

IP	Device	Zone	Bus	Breakers
10.0.0.5	RTU-3	ZONE-S	6	BR-5-6, BR-6-11, BR-6-12, BR-6-13

regulatory_compliance/compliance.md

CIP-008-6 R4: one hour after determination of a Reportable Incident; end of next calendar day for an attempt.

2 · Indexing

two paths, deliberately

seed

DETERMINISTIC

seed_graph.py

entities.yaml → 46 canonical nodes injected verbatim

extract

LLM EXTRACTION

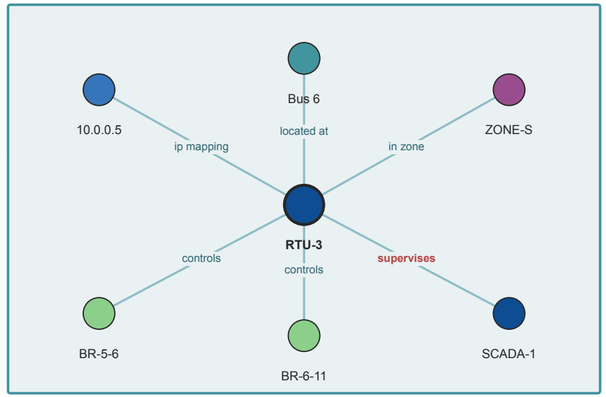
llama3.1:8b

107 concepts + relation descriptions

the 8B under-extracted IPs / buses / breakers, so the physical join is injected, not learned

3 · Knowledge graph

153 entities · 134 relations



"Loss of communication with an RTU means loss of visibility and control of its breakers" — carried on the EDGE, not on a node

4 · Retrieval

one query PER PERSONA (§11.1)

resolve_device(10.0.0.5)

table join · always first · never truncated

operator

which breakers, manual control, physical risk

data scientist

what the features mean, what they can't prove

regulator

reportable? which clock? notify whom?

one ranking could not serve three needs: the CIP rule reached the model in 0 / 30 blocks before this

5 · Role-specific narratives

tier C — the system under test

OPERATOR

Affected: RTU-3, IEC-104 station, 10.0.0.5:2404, ZONE-S at Bus 6.
Controls BR-5-6, BR-6-11, BR-6-12, BR-6-13.
Main risk is loss of view/control at Bus 6, not confirmed breaker operation.

REGULATOR

Reportable as an ATTEMPT under CIP-008-6 R4 — end of next calendar day to E-ISAC/NCCIC.
The detector does not establish compromise.

every claim traces back to a specific edge — see docs/notes/worked_example.md

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The two departures from stock LightRAG

① A deterministic path runs alongside LLM extraction. The 8B under-extracted IPs, buses and breakers, and the synthesizer then invented device names ("RTU-123"). Canonical entities are now injected from entities.yaml, and the IP→device join is a table lookup prepended to every context block — so the network→physical link is never left to semantic search. Coloured nodes above are seeded; grey concepts elsewhere in the graph are LLM-extracted.

② Retrieval is persona-conditioned, not shared. One query per alert, reused for three personas, ranked the compliance chunk last: the CIP reporting rule reached the synthesizer in 0 of 30 context blocks, and the model filled the gap with a "24 hours" figure found nowhere in the knowledge base. Three targeted queries put it in 30 of 30. The cost, stated plainly: three retrievals per alert instead of one.